

APPROVED

by Academic Council of
“Astana IT University” LLP
Minutes No. 8
dated April 30, 2025

**METHODOLOGICAL GUIDELINES FOR THE IMPLEMENTATION
OF MASTER’S RESEARCH WORK AND SCIENTIFIC INTERNSHIP
Of «ASTANA IT UNIVERSITY» LLP**

Astana, 2025

Terms and Definitions

Term	Definition
Final Assessment of Students	A procedure conducted to determine the extent to which students have mastered the scope of academic disciplines and/or modules, as well as other types of academic activities provided by the educational program, in accordance with the state compulsory education standard.
Academic Credit	A standardized unit of measurement of a doctoral student's academic workload; one credit equals one academic hour of classroom work per week during an academic period (trimester).
Master's Degree	An academic degree awarded to individuals who have completed master's degree programs.
Master's Thesis	The final project of a student enrolled in a research and pedagogical master's program. It represents an independent scientific study containing theoretical and/or practical developments on a relevant issue in the field of the chosen educational program, based on modern theoretical, methodological, and technological advances in science and technology.
Master's Research Work	Research work prescribed within the research and pedagogical master's program, including the completion of the master's thesis.
Letter-Grade Point System of Assessing Academic Achievement	A system for assessing the level of students' academic performance in points corresponding to the internationally recognized letter grading scale with a numeric equivalent, allowing for the determination of students' ranking.
Educational Program (hereinafter-EP)	An integrated set of the main characteristics of education, including the goals, learning outcomes, and content of studies, the organization of the educational process, methods of delivery, and criteria for evaluating learning outcomes.
Scientific Novelty of the Master's Thesis	New knowledge not previously known and formulated in the course of thesis preparation, substantiated from a theoretical perspective or as a result of experiments, tested in practice, and confirmed by the candidate.

4. 1. General Provisions

1. The Methodological Guidelines for the Implementation of Master's Research Work at "Astana IT University" LLP (hereinafter – the Guidelines) define the requirements for the content of the master's thesis, as well as for the performance of research work and internship by master's students at "Astana IT University" LLP (hereinafter – AITU), in accordance with Order No. 2 of the Ministry of Science and Higher Education of the Republic of Kazakhstan dated July 20, 2022, "On the Approval of State Compulsory Standards of Higher and Postgraduate Education."

2. Requirements for the Master's Student's Research Work:

- must correspond to the profile of the master's degree program under which the master's thesis is defended;
- must be relevant, contain scientific novelty, and have practical significance;
- must be based on modern theoretical, methodological, and technological achievements in science and practice;
- must employ modern research methodology and methods of data processing and interpretation;
- must contain theoretical (methodological, practical) sections consistent with the scientific provisions defended in the thesis;
- may be carried out within the framework of funded fundamental, applied, state budget, contractual, and initiative-based projects.

2. Organization of Master's Research Work

3. The master's student's research work includes:

- Master's student's research work (MSRW);
- scientific internship at research institutions and/or organizations of relevant industries or fields of activity;
- scientific publications or presentations at scientific and practical conferences with published abstracts;
- writing and defense of the master's thesis.

4. The master's student, together with the academic supervisor, develops an Individual Work Plan (IWP) of the master's student, which includes disciplines based on the educational program (EP), and it must be approved no later than three months after enrollment (Appendix 1).

5. Amendments to the IWP are made on the basis of a substantiated application by the master's student and approved by the Director of the EP School in coordination with the Office of Postgraduate Education.

6. As part of the implementation of the MSRW, the academic supervisor prepares a consultation schedule.

7. The MSRW is carried out in accordance with the academic calendar, curriculum, and IWP. Reports on the MSRW are prepared by the master's student in

accordance with the requirements of the EP School and uploaded to the LMS Moodle system. A sample title page of the MSRW report is provided in Appendix 2.

8. The academic supervisor evaluates the performance indicators of the MSRW using the letter-grade point system and enters the grade into the LMS Moodle system.

9. Based on the results of the MSRW, master's students, depending on their training profile, may perform the experimental part of their work at research institutes and scientific organizations that have signed the relevant agreements, during the period of scientific internship or research practice.

10. The main results of the master's thesis must be presented in at least one publication and/or one presentation at a scientific and practical conference with published abstracts.

11. To avoid publications in questionable outlets and predatory journals, it is recommended that master's students publish the results of their research in AITU editions and participate in conferences organized or co-organized by AITU. In the case of publications in other editions or participation in conferences, the master's thesis supervisor must be listed as a co-author of the article or provide written consent for such publication.

12. The final outcome of the master's student's research work is the master's thesis.

13. The requirements for the content, formatting, and defense of the master's thesis are defined in the Methodological Guidelines for the Completion of Master's Theses and Projects at "Astana IT University" LLP (Minutes No. 6 dated February 28, 2023).

3. Procedure for Conducting the Research Progress Evaluation of Master's Students

14. Each year, upon completion of the academic year, the master's student undergoes a Research Progress Evaluation to assess the implementation of the Individual Work Plan (IWP), including the Master's Research Work (MRW).

15. To the Research Progress Evaluation, a Research Progress Evaluation Committee is formed within the School of the Educational Program (EP School) to assess the fulfillment of the IWP by the master's student. The committee consists of faculty members of the respective school who hold an academic degree of Candidate of Sciences, Doctor of Sciences, or PhD in the relevant field of training. The composition of the committee (no more than five members) and its work schedule are organized by the EP School prior to the evaluation. The composition of the Research Progress Evaluation Committee is approved by the Rector of the University for one academic year.

16. The results of the Research Progress Evaluation Committee's work are discussed at a meeting of the EP School, and the decision is recorded in the minutes of the meeting in the form of a conclusion. The evaluation is conducted using the format "Passed" or "Not Passed".

17. The Director of the EP School submits an order to the Department of Academic Affairs with information on master's students who did not pass the Research Progress Evaluation for dismissal.

18. To be admitted to the final evaluation, the master's student must fully complete 100% of the plan in each section of the IWP.

19. Master's students who, after the first year of study, are not evaluated as having completed the IWP remain for the summer semester.

20. The master's student submits a report on the implementation of the IWP to the Research Progress Evaluation Committee for the Research Progress Evaluation.

4. Scientific Internship of Master's Students

22. As part of the Master's Research Work (MRW), to become familiar with innovative technologies, master's students are required to complete a scientific internship. Scientific internships provide students with the opportunity to apply research knowledge, master modern working methods, gain international experience, and establish academic connections. They may take place both in the Republic of Kazakhstan and abroad.

23. The place of the scientific internship must correspond to the scientific field of the educational program and the topic of the master's thesis.

24. The scientific internship is an integral part of postgraduate educational programs. Failure to complete the internship disqualifies the student from the final assessment.

25. The duration of the scientific internship for students of the scientific-pedagogical track is limited to no more than once during the entire period of study and must last at least 14 days.

26. A master's student may select the internship base (program) from the list proposed by the EP School, within the framework of agreements concluded by AITU with organizations designated as internship bases, or agreements on international internships according to the training direction. Alternatively, the student may propose their own internship base. In this case, the proposed base (program) is reviewed by the EP School for compliance with the internship requirements.

27. The internship program is developed jointly with the host organization. Both the internship program and the weekly schedule must be approved. A template of the program and schedule for the scientific internship is provided in Appendix 3. The master's student is responsible for fulfilling the internship program plan.

28. In the case of a group scientific internship, a joint internship program is developed (Appendix 4).

29. The EP School submits a memorandum to the Office of Postgraduate Education addressed to the Vice-Rector in charge, based on the required package of documents (agreement, invitation letter (if available), scientific internship program, internship plan, and application (Appendix 5)) in order to issue an order for assigning students to scientific internships at enterprises, organizations, institutions, universities, and research centers in Kazakhstan in relevant sectors or fields of activity.

30. A scientific internship at a partner organization may be conducted either free of charge or with a participation fee not exceeding the amount allocated in the current year's budget per student.

International scientific internships are implemented on a competitive basis. If extra-budgetary funds are available, the University covers the partner organization's service fees as well as accommodation, travel, and daily expenses, within the limits approved for the current year per student. Maximum reimbursement limits for international internships of AITU master's students are provided in Appendix 6.

31. By order of the AITU Rector, a Commission for the Selection of Candidates is established to select students for international scientific internships (hereinafter referred to as the Commission).

32. Master's students without academic debts are eligible to participate in international scientific internships.

The main criteria for competitive selection are:

1) Completion of one academic period at AITU with academic performance of "A," "A-," "B+," "B," or "B-";

2) Possession of a language proficiency certificate:

◦English:

- TOEFL ITP – minimum score 163;
- TOEFL iBT – minimum score 60;
- TOEFL PBT – minimum score 498;
- TOEFL PDT – minimum score 65;
- IELTS – minimum score 6.0;

◦German:

- DSH, Level C1;
- TestDaF, Level C1;

◦French:

- TFI – not lower than Level B1 in reading and listening;
- DELF, Level B2;
- DALF, Level C1;
- TCF – minimum score 50.

34. For the organization of an international internship, an invitation letter from the partner institution specifying the period of stay is required.

35. In the case of a master's student independently initiating an international internship at their own expense, item 30 does not apply. The student is reimbursed only for consular fees, within the limit approved for the current year per student.

36. No later than 60 calendar days before the international internship, the master's student must submit the following documents to the EP School:

- Application addressed to the Rector with endorsements from the academic advisor and EP School Director (Appendix 5);
- Foreign language certificate (if required);
- Translation into Kazakh or Russian of the invitation letter with a copy of the original;
- Internship plan agreed in writing with the international partner organization, prepared according to the form in Appendix 3.
- Within 10 (ten) days after the end of the internship, the student must submit an internship report (Appendix 7), as well as one of the following documents confirming completion:
 - Internship completion certificate;
 - Internship confirmation letter;
 - Reference (recommendation letter) from the internship host supervisor;
 - Diploma or certificate of an international program.

Additionally, in the case of paid services provided by the partner organization, the following documents must be submitted:

- Accommodation: invoice (tax) or receipt showing payment, rental agreement, act of completed work;
- Transportation: electronic flight ticket and boarding pass for each route; train tickets; bus tickets with dispatcher's reference (if available). For cash payments, a fiscal receipt is mandatory;

- Confirmation of paid internship services (online/offline): tripartite agreement, act of completed work, invoice, and payment documents confirming receipt and payment of the service.

37. The documents confirming the completion of the internship are reviewed and approved at a meeting of the EP School.

*Template of the Master's Student's Individual Work Plan***Astana IT University
School of EP****“Approved”
Director of the School of EP**_____

(signature, full name)

«__» _____ 20__

**“Approved”
Head of the Office of
Postgraduate Education**_____

(signature, full name)

«__» _____ 20__

MASTER'S STUDENT INDIVIDUAL WORK PLAN_____
(Full Name of the Master's Student)

Enrollment Order No. _____ dated _____

Program: Scientific and Pedagogical

Code and Title of the Educational Program _____

Period of Master's Studies: 20__-20__

Research Supervisor _____
(Full Name, Academic Degree, Academic Title)

Master's Thesis Topic::

in Kazakh: «_____»;

in English: «_____»;

in Russian: «_____»;

Approved by the AITU Academic Council, Minutes No. ____ dated «_» _____ 20__

Submission Deadline for the Master's Thesis: «_____» _____ 20__

Academic Year 20__ – 20__

Core Courses (CC) – _____ number of credits

[illegible][illegible]**Profiled Courses (PC) _____ number of credits**[illegible]

2) Elective Component (PC EC) _____ number of credits

Course Code	Course Title	Trimester	Number of Credits	Number of Weeks	Contact Hours	IWDS	Independent Work with Research Supervisor (IWRs)	Forms of Assessment

2. Practical Training __ number of credits

	Title	Trimester	Number of Weeks	Number of Credits	Host Organization
1	Pedagogical				
2	Research				
3	Presentation of Results (Report)				

Master's student _____ "_____" _____ 20____
(signature)

3. Research proposal

Master's Thesis Topic: «_____»

1. Relevance of the research
2. Scientific novelty
3. Research objectives and tasks
4. Research materials and methods
5. Literature review of the research topic
6. Timeline for the completion of the thesis work

No	Work content / Scope of work	Timeline (month, year)	Expected outcomes
----	------------------------------	------------------------	-------------------

	1-st year		
1			
2			
3			
	2-nd year		
4			
5			
6			

Master's student _____ «____» _____ 20____
(signature)

Research supervisor _____ «____» _____ 20____
(signature)

4. Completion of Research Internships

No	Location	Duration of internship	Outcomes	Completion mark
	Host organization / Country			

5. Publication Plan

№	Title of article / thesis	Name of journal / conference	Timeline	Completion mark

6. Research Work Plan (according to the academic calendar)

№	Type of work	Period of implementation	Form of reporting
1	MSRW 1 (describe the content of the work according to the approved topic)	According to the academic calendar	<i>Report to the academic supervisor (MSRW 1)</i>
2	MSRW 2 (describe the content of the work according to the approved topic)		
3	MSRW 3 (describe the content of the work according to the approved topic)		

4	MSRW 4 (describe the content of the work according to the approved topic)		
5	MSRW 5 (describe the content of the work according to the approved topic)		

7. Participation in a Research Project (if applicable)

Project registration number (IRN), Project title	Implemented within PEF / GF / international project / contract-based work	Full name of academic supervisor, position, degree	Implementation period	Position in the project

Research Supervisor:

Full name _____ «_____» _____ 202_

Master's Student:

Full name _____ «_____» _____ 202_

Title Page of the Master's Scientific Research Work Report

Astana IT University
School of _____ (educational program)

Approved
Director of School of _____ (name of EP)

(signature, full name)

Discussed at the meeting of the School of _____ (name of Educational Program)

Minutes № _____

«_____» _____ 20____

**REPORT
ON RESEARCH WORK
FOR the period of _____**
(period)

Master's student _____
(full name)

Educational program _____
(code, name)

Duration: from 20____ to 20____

Topic of doctoral thesis:

In Kazakh: «_____»;

In Russian: «_____»;

In Russian: «_____»;

The topic of the master's thesis was approved by the Academic Council, (No. and date of the minutes)

Deadline for submission of master's thesis «_____» _____ 20____

Research Supervisor:

Full name _____ «_____» _____ 202_

Master's Student:

Full name _____ «_____» _____ 202_

Astana, 20____

APPROVE

Vice-Rector for Science and Innovations

“Astana IT University” LLP

_____ Full Name

signature

« _____ » _____ 20__

APPROVE

Official of the host organization

Name of the host organization

_____ Full Name

signature

« _____ » _____ 20__

Master's Student Scientific Internship Program*FULL NAME*

Educational Program: _____

code, name

Year: _____

Master's thesis topic: _____

Scientific supervisor: _____

Location of the research internship (organization, country, city) _____

Internship period _____

Internship program content:**I. Educational component**

1. Topic 1
2. Topic 2
3. ...

II. Research component

1. Task 1
2. Task 2
3. ...”

Weekly internship plan

Date	Time	Description of tasks	Responsible person

APPROVED:

Scientific supervisor _____ FULL NAME

Scientific supervisor from host organization _____ FULL NAME

Director of the Academic Program School _____ FULL NAME

APPROVE

Vice-Rector for Science and Innovations

“Astana IT University” LLP

_____ FULL NAME

signature

«_____» _____ 20__

APPROVE

official representative of the host organization

name of the host organization

_____ FULL NAME

signature

«_____» _____ 20__

Scientific Internship Program on the topic:**Internship Period:** _____**Content of the Internship Program:****I. Educational Component**

1. Topic 1
2. Topic 2
3. Topic 3
4. ...

II. Research Component

1. Task 1
2. Task 2
3. Task 3
4. ...

Weekly Internship Plan

Date	Time	Description of Work	Person in charge
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APPROVED:

Director of the School of

Educational Programs _____ FULL NAME

Director of the Department of

Science and Innovation _____ FULL NAME

Appendix 5

*Form of Application for Sending a Master's
Student for Internship*

To the Director of the School

(name)

FULL NAME

from a master's student of the ____ year of
study of the educational program

(code, title)

FULL NAME

studying on a grant / paid basis

mobile phone number

Application

I kindly request permission to be sent for a research internship
at _____

((name of the organization, country, city))

on the topic of my master's thesis: _____

_____ for the period from _____ to _____ on an unpaid basis
(and to cover the cost of the internship in the amount of _____ tenge).

Signature
Date

APPROVED:

Scientific supervisor _____ FULL NAME

Director of the School _____ FULL NAME

**Reimbursement Rates for Expenses during International Research
Internships of AITU Master's Students**

№	Type of Expense	Rates Outside the Republic of Kazakhstan
1	Daily allowance	8 MCI (Monthly Calculation Index) per day
2	Travel	Based on actual expenses, supported by documentation
3	Accommodation	Based on actual expenses, supported by documentation
4	Visa processing and medical insurance	Based on actual expenses, supported by documentation
	Total	Maximum reimbursable amount – 600,000 tenge

«Astana IT University» LLP

*school***APPROVED**

Director _____

school

_____ FULL NAME

«_____» _____ 20__

Discussed at the meeting

Minutes No. _____

dated “” _____ 20__

**REPORT
ON RESEARCH INTERNSHIP**

Master's student

FULL NAME, group

Educational Program:: _____

title

Duration of the Master's Program: from “” _____ 20 to “” _____ 20

Year of Study: _____

Master's Thesis Topic: _____

Scientific Supervisor: _____

Place of Research Internship (organization, country, city): _____

Internship Period: from “” _____ 20_ to “” _____ 20_

Astana, 20__

Report on Research Internship

1. Types of Work Performed According to the Research Internship Plan and the Significance of the Results Obtained

№	Description of Work	Results Obtained

2. Participation in Conferences, Seminars, Round Tables, Academic Courses, etc. within the Framework of the Research Internship (if applicable)

№	Event Title	Results

3. Publications Based on the Results of the Research Internship (if applicable)

№	Article Title	Publisher, Place of Publication	Number of Printed Pages

4. Master's Student Feedback on the Effectiveness and Impact of the Research Internship

Master's Student _____
(signature, Full Name)

Scientific Supervisor _____
(signature, Full Name)

Head of the Internship Base (Responsible Person) _____
(signature, Full Name)

“_____” _____ 20